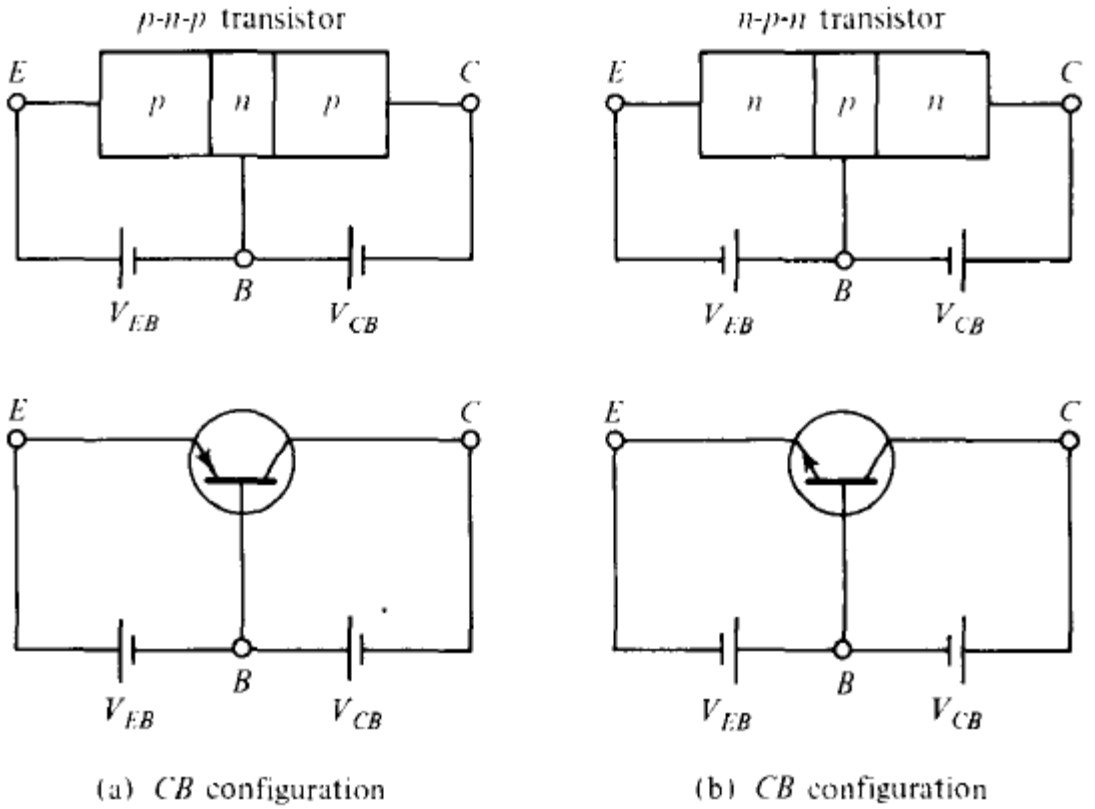
Transistor Configurations

- There are three types of circuit connections (called configurations) for operating a transistor. Each method of connection responds differently to its input signal.
- The term ‘common’ is used to denote the electrode that is common to the input and output circuits.
- Since a transistor is a 3-terminal (and not a 4-terminal) device, one of its terminals has to be common to the input and output circuits.
- The three transistor configurations include

1. Common Base Configuration - has voltage gain but no current gain.
2. Common Emitter Configuration - has both current and voltage gain.
3. Common Collector Configuration - has current gain but no voltage gain

Common Base Configuration

- The *common-base* (CB) *configuration* refers to the one where the base terminal is common to both the emitter (input circuit) and collector (output circuit) terminals as shown below



(a) *CB* configuration (b) *CB* configuration

Common Base Configuration

- The CB configuration is also called the *grounded-base configuration.*
- For a *p-n-p* transistor, the largest current components are caused by holes. Holes flow from the emitter to the collector and down toward ground out of the base terminal.
- In an *n-p-n* transistor all current and voltage polarities are negative to those in a *p-n-p* transistor.
- The CB configuration of a transistor is usually used in amplifier applications.

Common Base Configuration – Current Amplification Factor

- The common base current gain, α is defined as the ratio of d.c. collector current to d.c. emitter current:

$$\alpha = \frac{I_C}{I_E}$$

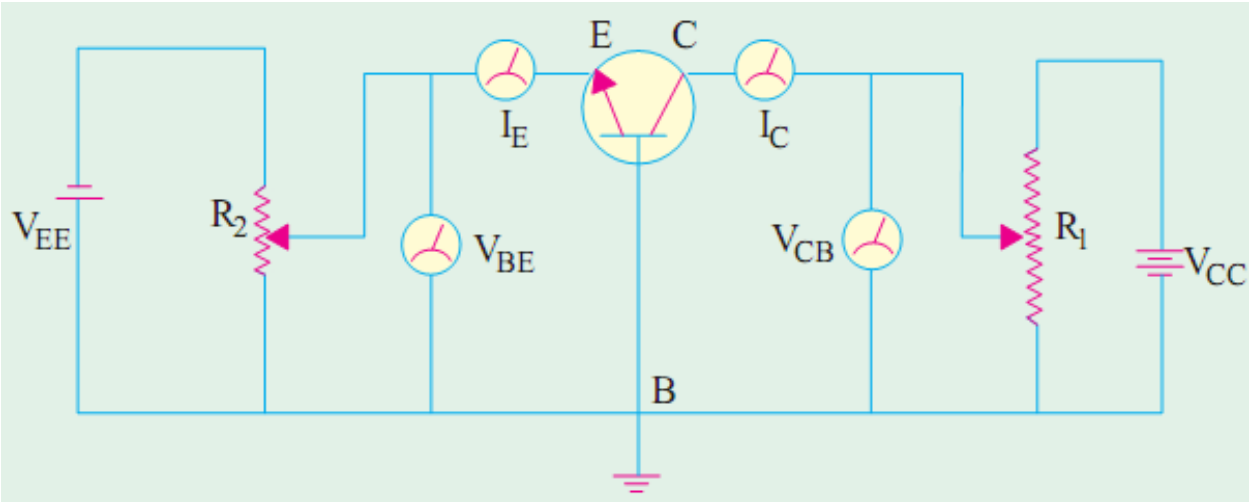
- The α of a transistor is a measure of the quality of a transistor; higher the value of α, better the transistor in the sense that collector current more closely equals the emitter current.
- Its value ranges from 0.95 to 0.999.
- From $I_E = I_B + I_C$

$$I_B = I_E - I_C = I_E - \alpha I_E$$

$$\Rightarrow I_B = I_E (1 - \alpha)$$

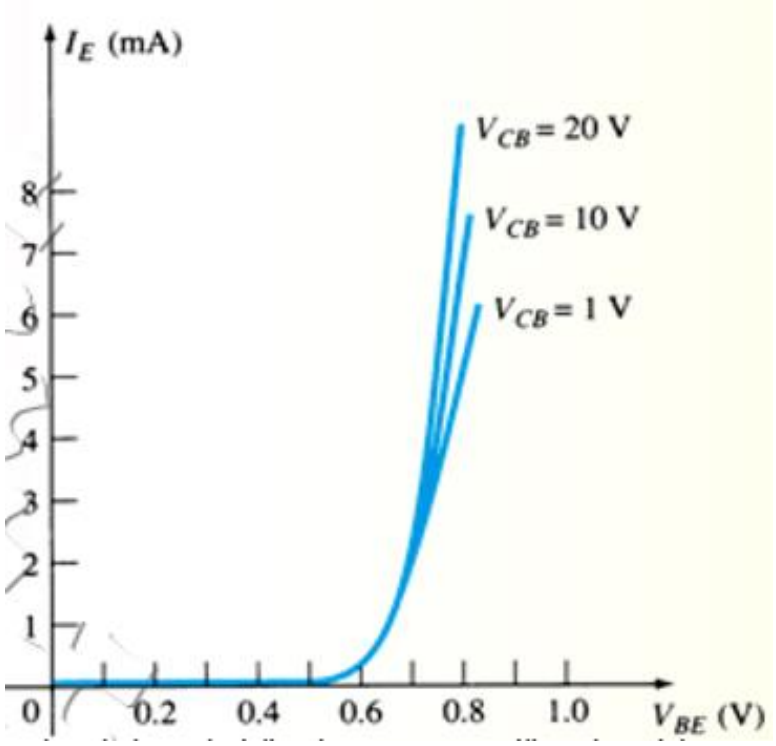
Common Base Configuration – Transistor Characteristics

- These are the curves which represents relationship between different dc currents and voltages of a transistor.
- There are two main transistor characteristics : **input** and **output characteristics**
- Transistor characteristic curves are helpful in studying the operation of a transistor when connected in a circuit.
- The static characteristics of an *NPN* transistor connected in common-base configuration can be determined by the use of test circuit shown below.
- Milliammeters are included in series with the emitter and collector circuits to measure I_E and I_C. Similarly, voltmeters are connected across E and B to measure voltage V_{BE} and across C and B to measure V_{CB}. The two potentiometer resistors R_1 and R_2 supply variable voltages from the collector and emitter dc supplies respectively.



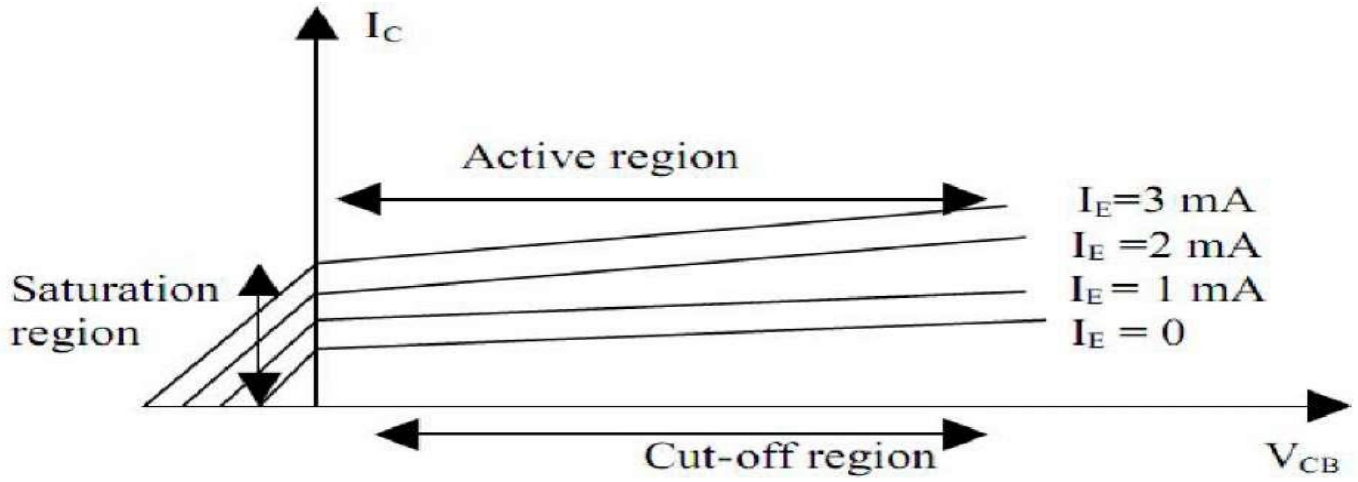
Common Base Configuration – Input Characteristics

- The input characteristic is a curve between I_E and emitter base voltage V_{BE}, keeping V_{CB} constant.
- I_E is taken along y-axis and V_{BE} is taken along x-axis. This is called the *input characteristic* because the base-emitter will become the *input terminals.*
- First, voltage V_{CB} is adjusted to a suitable value with R_1. Next, voltage V_{BE} is increased in a number of discrete steps and corresponding values of I_E are noted from the milliammeter connected for the purpose. When plotted, we get the input characteristic curve.



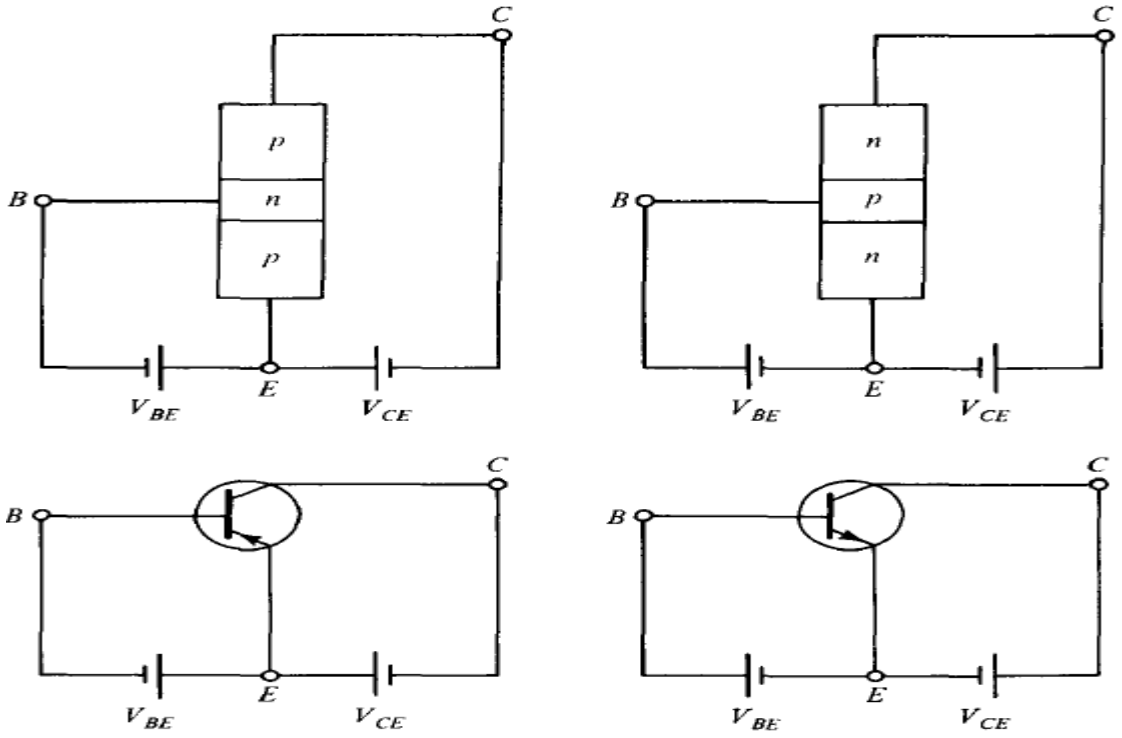
Common Base Configuration – Output Characteristics

- The output characteristic is a curve between I_C and V_{CB} at constant I_E. The collector current I_C is taken along y-axis and V_{CB} is taken along x-axis .
- The method of obtaining this characteristic is as follows:
  - First, movable contact, on R_2 is changed to get a suitable value of V_{BE} and hence that of I_E. While keeping I_E constant at this value, V_{CB} is increased from zero in a number of steps and the corresponding collector current I_C that flows is noted.
  - Next, V_{CB} is reduced back to zero, I_E is increased to a value a little higher than before and the whole procedure is repeated. In this way, whole family of curves is obtained, a typical family being shown in figure below.



Common Emitter Configuration

- Most transistors have their emitter, rather than their base, as the terminal common to both input and output networks. Such a configuration is known as common-emitter (CE) or grounded-emitter configuration as shown below



Common Emitter Configuration

- The CE configuration is commonly used as a switch or pulse transistor amplifier.
- This is because the transistor is open at the cutoff mode and is closed at the saturation mode.

Common Emitter Configuration - Current Amplification Factor

- The current amplification factor of the common emitter configuration , β_{dc} is the ratio of the dc collector current to the dc base current :

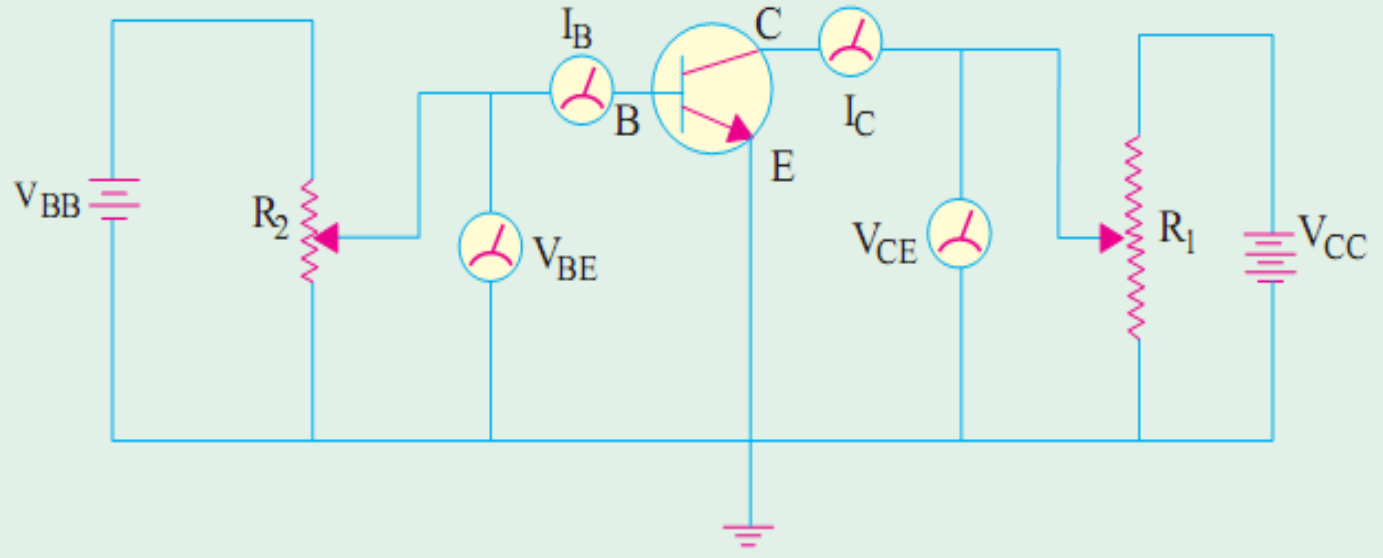
$$\beta_{dc} = \frac{I_C}{I_B}$$

- The dc beta of a BJT is related to the current gain alpha, by

$$\beta_{dc} = \frac{\alpha}{1 - \alpha}$$

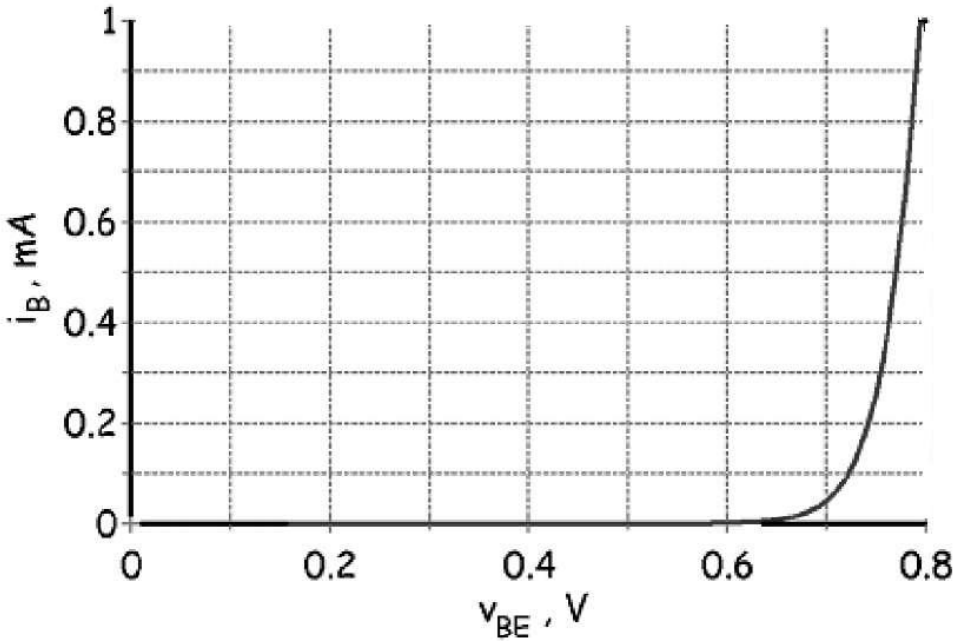
Common Emitter Configuration - Transistor Characteristics

- The static characteristics of a transistor connected in *CE* configuration may be determined by the use of circuit diagram shown in the figure below.
- A milliammeter is connected in series with the base to measure I_B. Similarly, a milliammeter is included in the collector circuit to measure I_C.
- A voltmeter with a typical range of 0 –1 V is connected across base and emitter terminals for measuring V_{BE}. Potentiometer R_2 connected across dc supply V_{BB} is used to vary I_B and V_{BE}.
- A second voltmeter with a typical range of 0–20 V is connected across collector-emitter terminals to measure the output collector-emitter voltage V_{CE}.



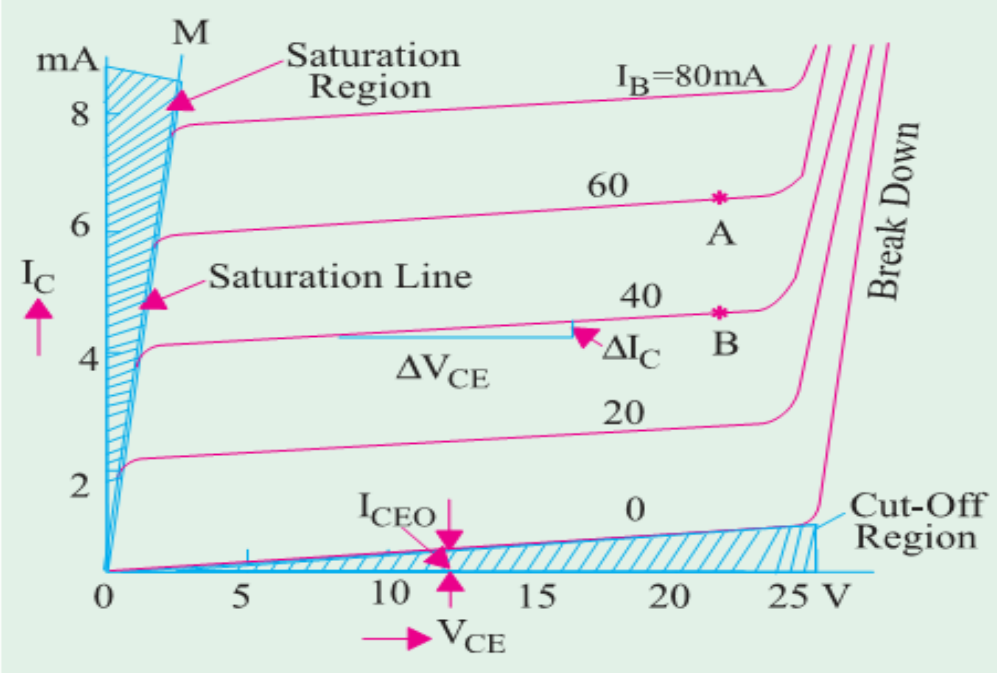
Common Emitter Configuration - Input Characteristics

- Input characteristic is a curve between V_{BE} and base current I_B at constant V_{CE}
- To begin with, voltage V_{CE} is maintained constant at a convenient value and then V_{BE} is increased in steps. Corresponding values of I_B are noted at each step.
- The procedure is then repeated for a different but constant value of V_{CE}. A typical input characteristic is shown in figure below



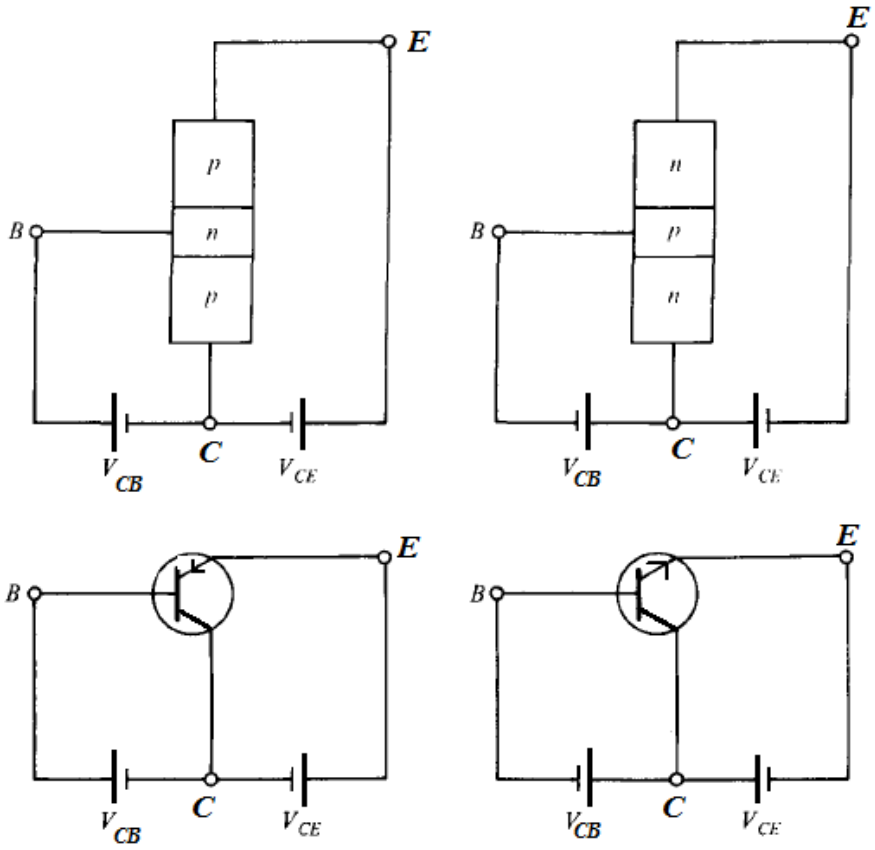
Common Emitter Configuration - Output Characteristic Curve

- It is the curve between V_{CE} and I_C at constant I_B
- For obtaining this characteristic, first I_B is set to a convenient value and maintained constant and then V_{CE} is increased from zero in steps, I_C being noted at each step.
- Next, V_{CE} is reduced to zero and I_B increased to another convenient value and the whole procedure repeated.
- In this way, a family of curves is obtained.



Common Collector Configuration

- Another transistor-circuit configuration is called the common-collector or grounded-collector configuration as shown below
- In a common-collector (CC) configuration, the output voltage of the load is taken from the emitter terminal instead of the collector as in the common-base and common-emitter configurations.



Common Collector Configuration

- When the transistor is cut off, no current will flow in the emitter terminal at the load.
- When the transistor is operating in a saturation mode, the load current reaches toward its maximum.
- Therefore the CC configuration transistor can also be used as a switch or pulse amplifier.
- It is used primarily for impedance-matching purpose since it has high input impedance and low output impedance.
- The significant difference, however, between common-emitter and common-collector configurations is that the common-collector amplifier has no voltage gain.

Common Collector Configuration – Current Amplification Factor

- The common collector current amplification factor, γ is

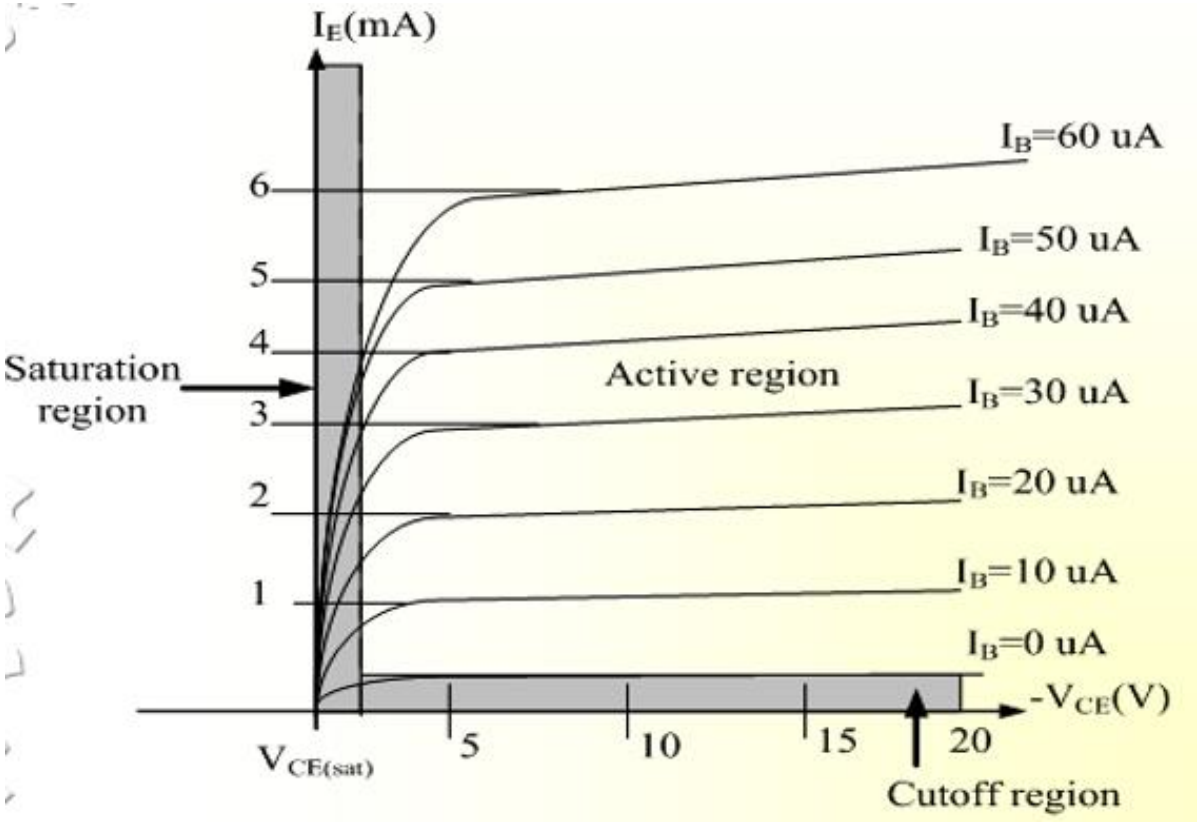
$$\gamma = \frac{I_E}{I_B}$$

- By substitution, it can be shown that,

$$\gamma = \frac{1}{1 - \alpha}$$

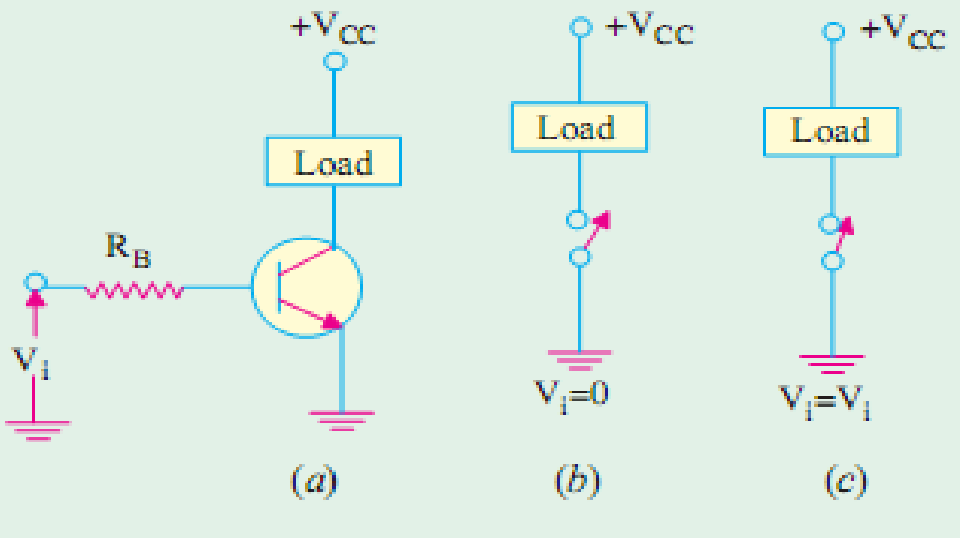
Common Collector Configuration - Output Characteristics

- For the common-collector configuration, the output characteristics are a plot of I_E vs V_{CE} for a range of values of I_B.



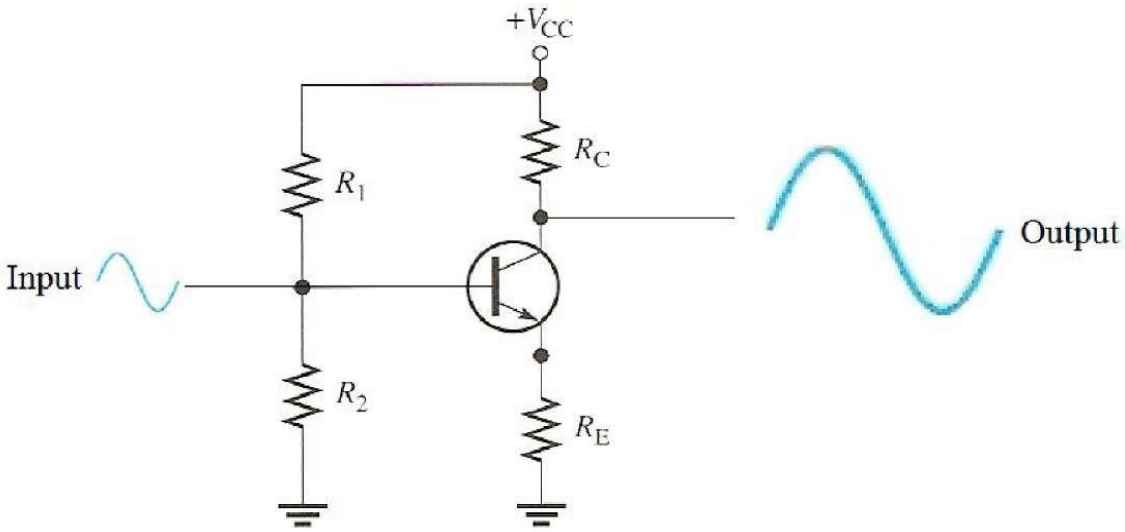
BJT as a Switch

- BJT can be made to operate as a switch.
- If the circuit uses the transistor as a switch, then the biasing is arranged to operate in saturation (ON) and cut-off (OFF).
- When using BJT as a switch, usually two levels of control signal are employed.
- With one level, the transistor operates in the cut-off region (open) whereas with the other level, it operates in the saturation region and acts as a short circuit.
- Fig. (*b*) shows the condition when control signal $v_i = 0$. In this case, the *BE* junction is reverse-biased and the transistor is open and, hence acts as an open switch.
- However, as shown in Fig. (*c*) if v_i equals a positive voltage of sufficient magnitude to produce saturation, the transistor acts as a closed switch.



BJT as an Amplifier

- In order to use a transistor as an amplifier, it should be operated in active region i.e. emitter junction should be always forward bias and collector junction should be reversed bias.
- Therefore in addition to the ac input *source* V_{in}, two *dc* voltages V_{BE} and V_{CE} are applied as shown in figure below. This *dc* voltage is called bias voltage.
- As the input circuit has low resistance, a small change in the input signal causes a change in the base current thereby causing a large change in collector current (because $I_C = \beta I_B$). The collector current flowing through the high load resistance, R_C, produces a large voltage across it. Thus a weak signal applied at the input circuit appears in the amplified form at the output.



DC Load Line of a BJT

- DC biasing is used to establish a steady level of transistor current and voltage called the dc operating point or quiescent point (Q-point)
- Consider a CE amplifier along with the output characteristics as shown in the figure below.
- A straight line drawn on the output characteristic of a transistor which gives the various zero signal values (i.e. when no signal applied) of V_{CE} and I_C is called ***DC load line***.

CE Amplifier

